

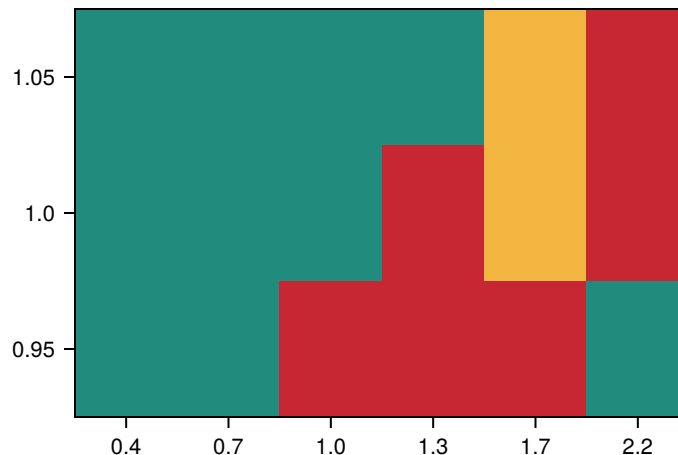
REAL IEEE-123 impedances — NO structural inexactness found

123 buses · public OpenDSS positive-sequence (Fortescue-reduced) impedances · 85 load nodes · T=24 · priced export ON · free cone-gap detector, no AC oracle
axes are multipliers on the Phase-17-retuned point (load 0.05 / pv 0.12, seed 20260719) — ×1.0 is the thesis-reproduction point
RED CELLS ARE SOLVER NOISE, NOT RELAXATION GAPS: tightening tol_gap 1e-8→1e-10 collapses the worst ratio 4.76→0.0029 at an identical optimum

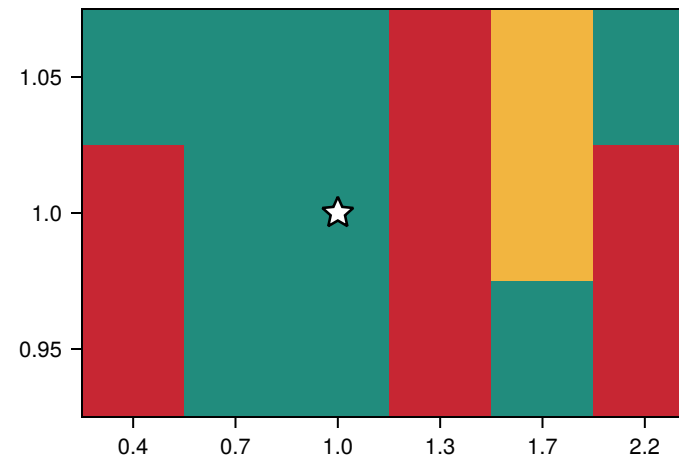
vmax = 1.05



vmax = 1.075



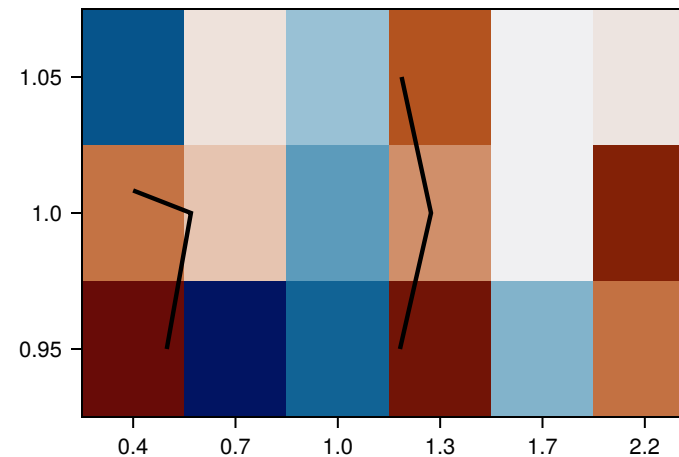
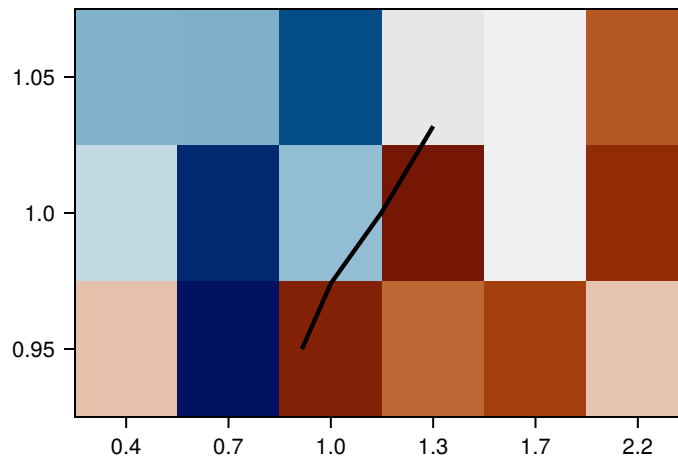
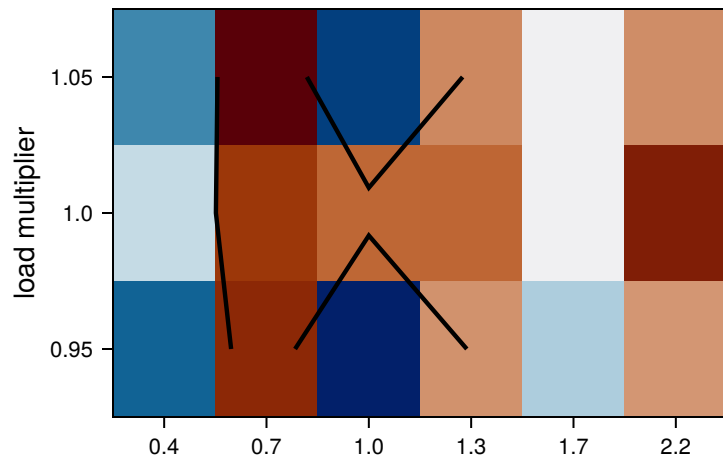
vmax = 1.1



pv multiplier

pv multiplier

pv multiplier



$\log_{10}$ cone-gap ratio

ratio ≤ 1 flagged ratio > 1 — SOLVER NOISE guard tripped — UNMEASURED not swept $\circ$ voltage bound active (never occurs) $\star$ thesis-reproduction point

Voltage upper bound is NEVER active at any of the 48 solved points (vpeak span 0.9997-1.0158) — the 3-bus overvoltage mechanism does not occur here.
Flags are scattered non-monotonically in every axis, including at the LOWEST pv swept ($\times 0.4$), which no physical relaxation-gap mechanism can produce.